1. Introduction to Probabilistic Reasoning

CSC3043S Reasoning in AI: Bayesian Networks

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[Based in part on slides created by Dan Klein and Pieter Abbeel for CS188 Intro to AI at UC Berkeley. All materials available at http://ai.berkeley.edu.]



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Uncertainty

A central problem in AI is dealing with uncertainty

- *Systematic uncertainty*: We only observe finite datasets – the more data, the better predictions can be made based on the data
- *Intrinsic uncertainty*: We can only observe partial information about the world, so we can't always model all the factors influencing a prediction

Probability theory is a framework for quantifying and reasoning about uncertainty

- It is more flexible than logic-based frameworks, and more principled than other approaches to handle degrees of truth such as fuzzy logic

Probabilistic models

A probabilistic model uses probabilities to describe the relationship between multiple variables that each have some degree of uncertainty

- The model's environment consists of:
  - **Observed variables (evidence)**: We know certain things about the state of the world (e.g., sensor readings or symptoms)
  - **Unobserved variables**: We need to reason about other aspects (e.g., where an object is or what disease is present)
  - **Model**: Some description about how the known and unknown variables are related to one other
- Models describe a how a (portion of) the world works
- Models are always simplifications
  - May not account for every variable or all interactions between variables
  - "All models are wrong; but some are useful." – George E. P. Box

Uncertainty

Probabilistic reasoning

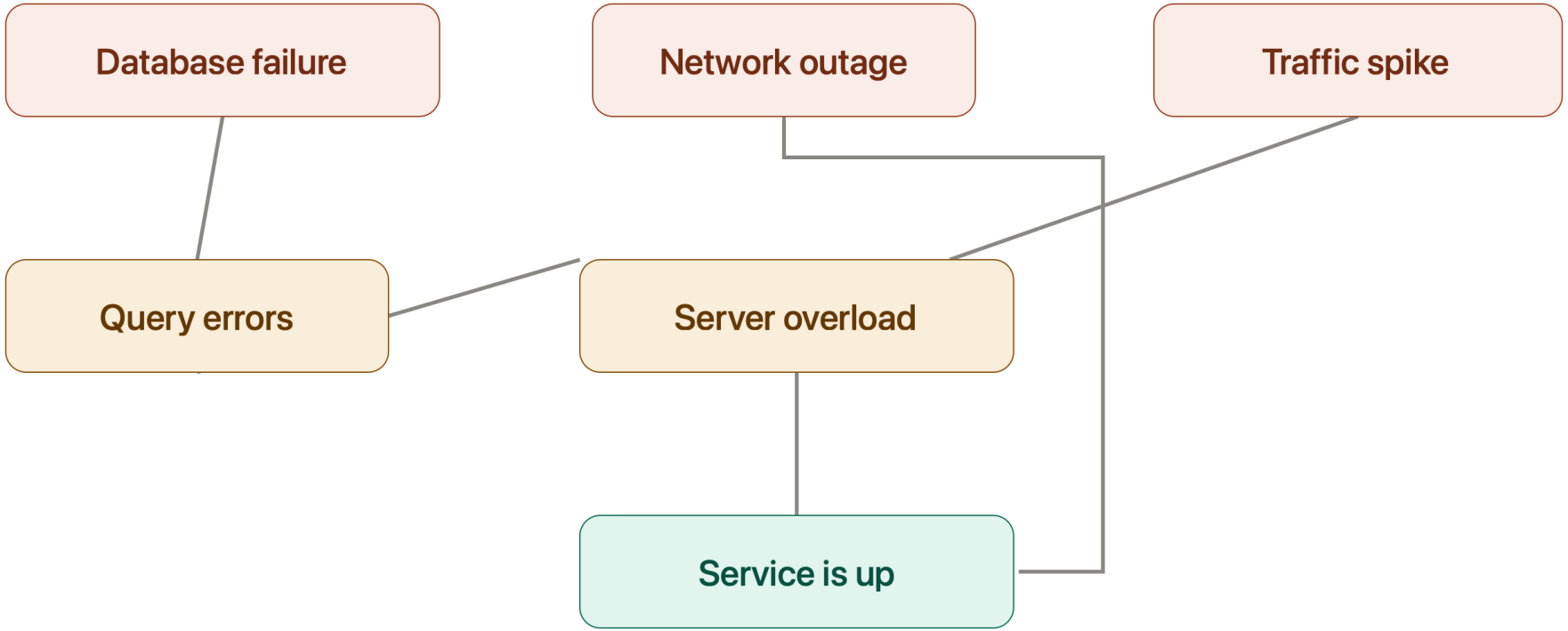
- Probabilistic reasoning provides a framework for managing our beliefs and knowledge (there may or may not be actual randomness/stochasticity)
- We (or our agents) use probabilistic models to to reason about unknown variables, given evidence
- We will see different kinds of reasoning, including *explanation* (diagnostic reasoning) and *prediction* (causal reasoning)

Deep learning model are powerful at making predictions of one variable given features as input, but can't reason directly about the relationship between different variables

Bayesian Networks

Framework for probabilistic models to represent knowledge and enable reasoning under uncertainty

- Example: web service fault diagnosis



Bayesian Networks

Lecture structure:

Date	Monday	Tuesday	Wednesday	Thursday	Friday
24-Aug		Probability	BN graphical models	BN inference	Tutorial
31-Aug	BN conditional independence	BN approx inference	DBNs/HMMs	Decision networks	Tutorial

Assignment: Closed practical on applying Bayesian networks

Lecture overview

- Random variables
- Probability distributions
- Events and marginal distributions
- Conditional probabilities
- Independence and conditional independence

Everything in this module builds on these concepts, so make sure you are familiar with them now

Random variables

- A random variable represents some aspect of the world about which we (may) have uncertainty, *e.g.*,
  - R = Is it raining?
  - T = Is it hot or cold?
  - D = How long will it take to drive to campus?

- Random variables have *domains,* e.g.,
  - R in {true, false} (can write as {+r, -r})
  - T in {hot, cold}
  - D in $[0, \infty)$

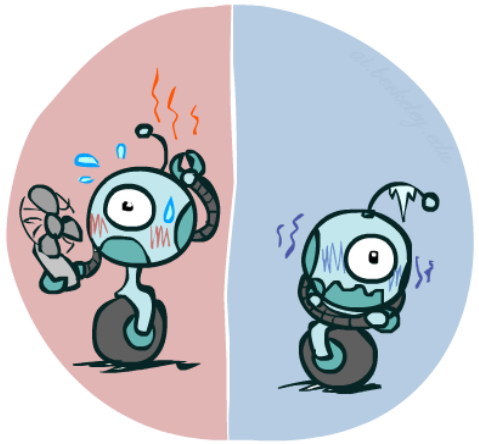
- Notation: We denote random variables with capital letters, and values or variables representing specific values in lower case

- In this course we will primarily work with discrete (categorical) variables

Probability Distributions

- Unobserved random variables have *distributions*
- A (discrete) **probability distribution** is a table of probability values
- Probability distributions associate each value in the domain with a probability

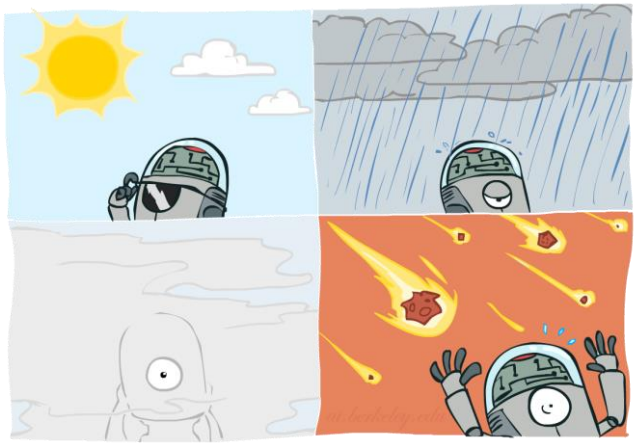
Temperature:



$P(T)$

T	P
hot	0.5
cold	0.5

Weather:



$P(W)$

W	P
sun	0.6
rain	0.1
fog	0.3
meteor	0.0

Probability Distributions

- Definition: A **probability** $P(X = x)$ is a single number with the following properties:

$$\forall x \;\; P(X = x) \geq 0$$

$$\sum_x P(X = x) = 1$$

Sum over the probabilities of all the values in the domain

- e.g. $P(W = rain) = 0.1$
- Shorthand notation (if all domain entries are unique):

$$P(rain) = P(W = rain)$$

Joint Distributions

A **joint distribution** over a *set* of random variables specifies a real number for each possible assignment to the variables.

- Example:

$$P(T, W)$$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Joint Distributions

- A **joint distribution** over a set of random variables $X_1, X_2, \ldots, X_n$ specifies a real number for each assignment. Notation:

$$P(X_1 = x_1, X_2 = x_2, \ldots X_n = x_n)$$

$$P(x_1, x_2, \ldots x_n)$$

- A joint distribution must have the following properties:

$$P(x_1, x_2, \ldots x_n) \geq 0$$

$$\sum_{(x_1, x_2, \ldots x_n)} P(x_1, x_2, \ldots x_n) = 1$$

Joint Distributions

- A **probabilistic model** is a joint distribution over a set of random variables
  - Assignments are also called *outcomes*
  - The joint distribution says how likely each outcome is

- The size (number of table entries) of a joint distribution over *n* variables with domain sizes *d* is d^n
  - We will need better ways of representing probabilistic models than large tables

Events

- An **event** is a set of outcomes *E*

$$P(E) = \sum_{(x_1 \ldots x_n) \in E} P(x_1 \ldots x_n)$$

- The probability of an event is sum over the probabilities of all outcomes consistent with the event
- Given a joint distribution we can calculate the probability of any event

Events

- Example: Calculate the probability of the following events:
  - Probability that it's hot AND sunny?
  - Probability that it's hot?
  - Probability that it's hot OR sunny?
- Typically, the events we care about are partial assignments, e.g., $P(T = hot)$

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Marginal distributions

- **Marginal distributions** are distributions over a subset of variables in a joint distribution
- **Marginalization**: Compute the marginal distribution by summing over the probabilities of all possible assignments of the variable(s) being marginalized

$$P(X_1 = x_1) = \sum_{x_2} P(X_1 = x_1, X_2 = x_2)$$

Marginal distributions

- Marginal distributions are *sub-tables that eliminate probabilities*: Combine collapsed rows by summing

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$$P(t) = \sum_w P(t, w)$$

$$P(w) = \sum_t P(t, w)$$

$P(T)$

T	P
hot	0.5
cold	0.5

$P(W)$

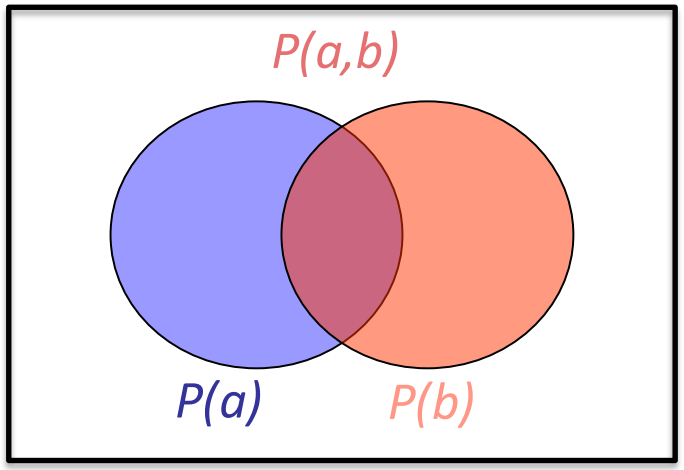
W	P
sun	0.6
rain	0.4

Conditional probability

- Definition of the **conditional probability** of *a* given *b:*

$$P(a|b) = \frac{P(a,b)}{P(b)}$$

- Intuition: What does observing *b* tell us about *a?*



Conditional probability

- Example:

P(sun|cold)

$$P(W = s|T = c) = \frac{P(W = s, T = c)}{P(T = c)} = \frac{0.2}{0.5} = 0.4$$

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

- The observed variable(s) are called the *evidence*

Conditional probability

- **Conditional distributions** are probability distributions over some variables given fixed values of others

Joint Distribution

$$P(T, W)$$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Full Conditional Distribution

$$P(W|T)$$

Conditional Distributions

$$P(W|T = hot)$$

W	P
sun	0.8
rain	0.2

$$P(W|T = cold)$$

W	P
sun	0.4
rain	0.6

Conditional probability

- Computing a conditional distribution

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$$P(W = s|T = c) = \frac{P(W = s, T = c)}{P(T = c)}$$
$$= \frac{P(W = s, T = c)}{P(W = s, T = c) + P(W = r, T = c)}$$
$$= \frac{0.2}{0.2 + 0.3} = 0.4$$

$P(W|T = c)$

W	P
sun	0.4
rain	0.6

Conditional probability

More efficient approach to compute conditional distribution:
the **normalization trick**

1. Select the joint probabilities matching the evidence
2. Normalize the probabilities in the selection
   i. Compute the sum over all entries: *Z*
   ii. Divide every entry by *Z*

$$P(X_1, X_2)$$
$$P(X_1, x_2)$$
$$Z = \sum_{x_1} P(x_1, x_2)$$

- The sum of the selection is the probability of the evidence

$$P(x_1|x_2) = \frac{P(x_1, x_2)}{P(x_2)} = \frac{P(x_1, x_2)}{\sum_{x_1} P(x_1, x_2)}$$

Conditional probability

- Example: Compute $P(W|T = c)$ using the normalization trick

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

SELECT the joint probabilities matching the evidence



$P(c, W)$

T	W	P
cold	sun	0.2
cold	rain	0.3

NORMALIZE the selection (make it sum to one)



$P(W|T = c)$

W	P
sun	0.4
rain	0.6

- Sum of the selection is the probability of the evidence $P(c)$:

$$P(w|c) = \frac{P(w, c)}{P(c)} = \frac{P(w, c)}{\sum_{w'} P(w', c)}$$

Product Rule

- Compute the joint distribution from conditional distributions
  - Rewrite definition of conditional probability

$$P(x,y) = P(y)P(x|y) \quad \Leftrightarrow \quad P(x|y) = \frac{P(x,y)}{P(y)}$$

- Example:

$P(W)$

R	P
sun	0.8
rain	0.2

$P(D|W)$

D	W	P
wet	sun	0.1
dry	sun	0.9
wet	rain	0.7
dry	rain	0.3

$\Leftrightarrow$

$P(D,W)$

D	W	P
wet	sun	0.08
dry	sun	0.72
wet	rain	0.14
dry	rain	0.06

Chain Rule

- More generally, we can write any joint distribution as an incremental product of conditional distributions

$$P(x_1, x_2, x_3) = P(x_1)P(x_2|x_1)P(x_3|x_1, x_2)$$

$$P(x_1, x_2, \ldots x_n) = \prod_i P(x_i|x_1 \ldots x_{i-1})$$

- But with a large number of variables the size of the conditional distribution tables will become unmanageable

Independence

- Definition: Two variables X and Y are ***independent*** if:

$$\forall x, y : P(x, y) = P(x)P(y)$$

- This says that the joint distribution over the two variables factors into a product of two simpler distributions
- Equivalent definition:

$$\forall x, y : P(x|y) = P(x)$$

- Notation: random variable X is independent of Y: $X \perp\!\!\!\perp Y$

Independence

Example: N fair, independent coin flips

- Marginal distribution of each variable:

$P(X_1)$

H	0.5
T	0.5

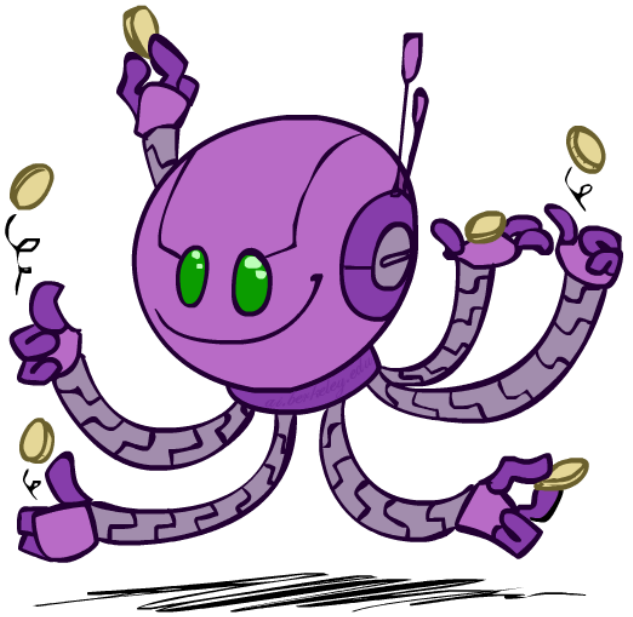
$P(X_2)$

H	0.5
T	0.5

. . .

$P(X_n)$

H	0.5
T	0.5



- Joint distribution $P(X_1, X_2, .. X_n)$ has 2^n entries

Independence

- Example: Are T and W independent?

$P_1(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Marginal distributions:

$P(T)$

T	P
hot	0.5
cold	0.5

$P(W)$

W	P
sun	0.6
rain	0.4

- Test if $P_1(t, w) = P(t)P(w)$ for all t, w
- Not the case here, so not independent

Independence

- Example: Are T and W independent?

$P_2(T, W)$

T	W	P
hot	sun	0.3
hot	rain	0.2
cold	sun	0.3
cold	rain	0.2

Marginal distributions:

$P(T)$

T	P
hot	0.5
cold	0.5

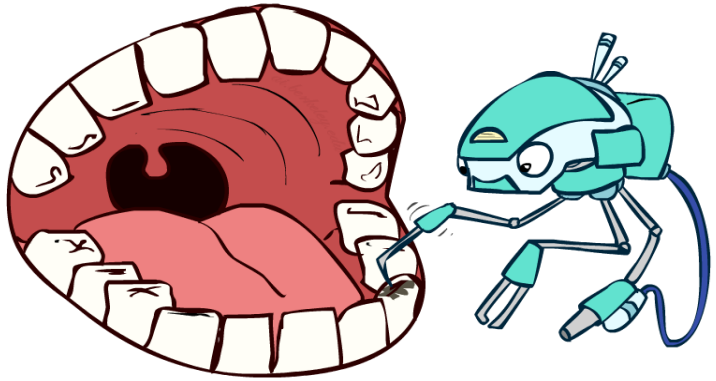
$P(W)$

W	P
sun	0.6
rain	0.4

- Test if $P_2(t, w) = P(t)P(w)$
- This holds for all assignments, so they are independent

Conditional Independence

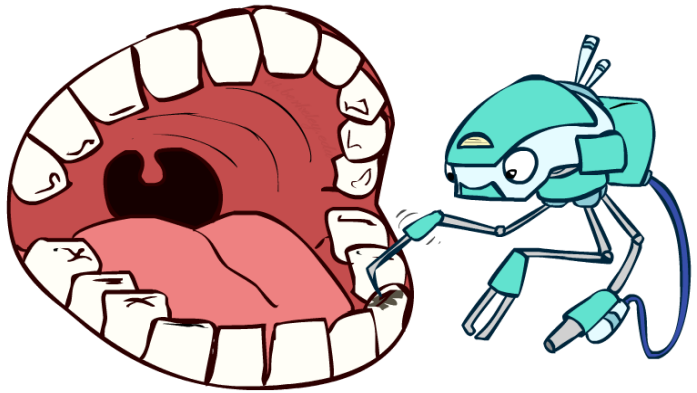
- In general, independence is a too strong assumption, but we will want to be able to express independence under certain conditions
- Example: Distribution over Boolean random variables: $P(Toothache, Cavity, Detect)$
- If one has a cavity, detection of the cavity doesn't depend on whether one has toothache:
  - $\mathrm{P}(+detect \mid +toothache, +cavity) = P(+detect \mid +cavity)$
- This also holds if there isn't a cavity:
  - $P(+detect \mid +toothache, -cavity) = P(+detect \mid -cavity)$



Conditional Independence

Example: Distribution over Boolean random variables:
$P(Toothache, Cavity, Detect)$

- We can say that Detect is ***conditionally independent*** of Toothache given Cavity:
  - $P(Detect \mid Toothache, Cavity) = P(Detect \mid Cavity)$
- Equivalent statements:
  - $P(Toothache \mid Detect, Cavity) = P(Toothache \mid Cavity)$
  - $P(Toothache, Detect \mid Cavity) = P(Toothache \mid Cavity)\, P(Detect \mid Cavity)$



Conditional Independence

- Unconditional (absolute) independence is very rare
- Conditional independence is our most basic and robust form of knowledge about uncertain environments
- Definition: X is **conditionally independent** of Y given Z if and only if:

$$\forall x, y, z : P(x, y|z) = P(x|z)P(y|z)$$

- Or equivalently, if and only if:

$$\forall x, y, z : P(x|z, y) = P(x|z)$$

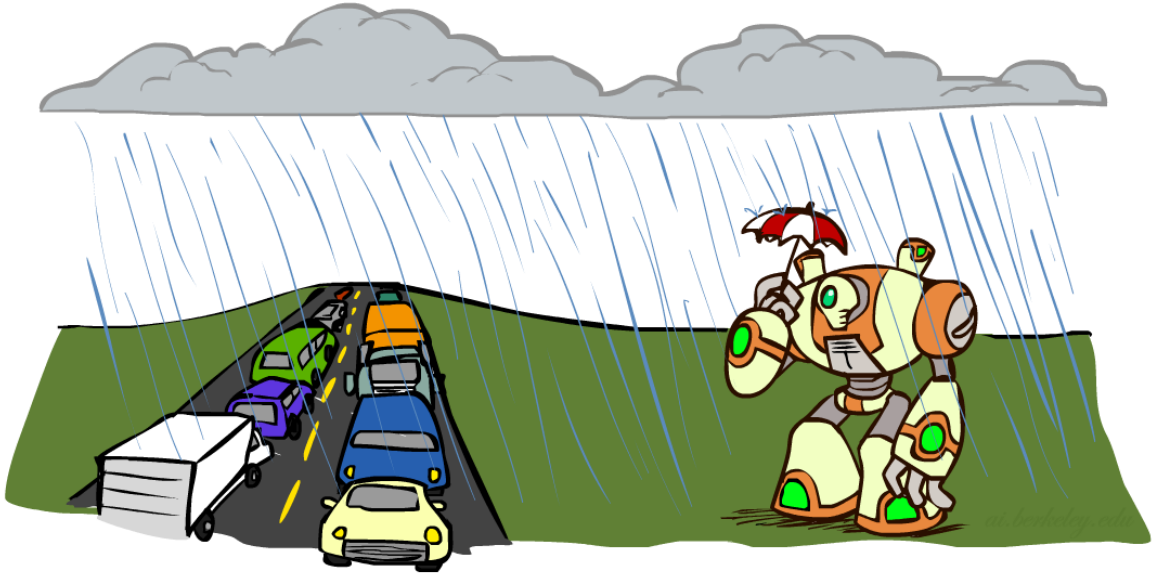
- Notation: $X \perp\!\!\!\perp Y|Z$

Conditional Independence

Example:

- Environment with random variables Traffic, Umbrella, and Raining

$$\text{Traffic} \perp\!\!\!\perp \text{Umbrella} | \text{Raining}$$

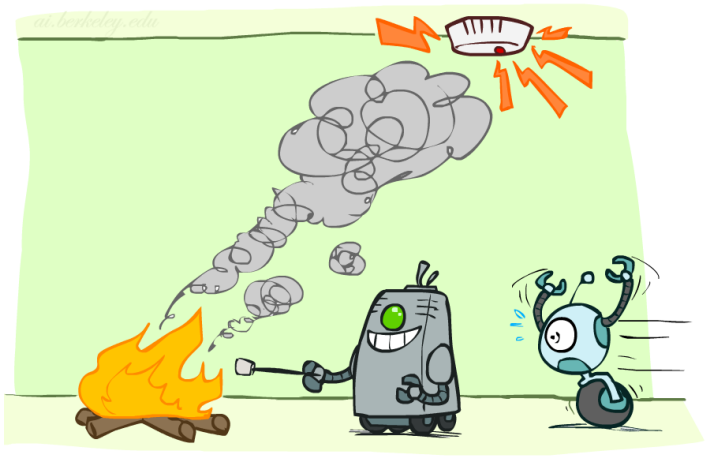


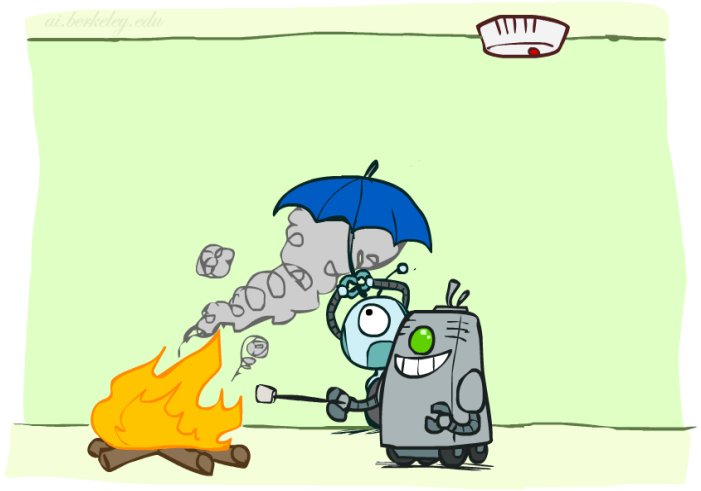
Conditional Independence

Example:

- Environment with random variables Fire, Smoke, and Alarm

$$\text{Alarm} \perp\!\!\!\perp \text{Fire} | \text{Smoke}$$





Conditional Independence

Conditional independence and the chain rule

- Chain rule: $P(X_1, X_2, \ldots X_n) = P(X_1)P(X_2|X_1)P(X_3|X_1, X_2)\ldots$
- Example: $P(\text{Traffic}, \text{Rain}, \text{Umbrella}) =$

  $P(\text{Rain})P(\text{Traffic}|\text{Rain})P(\text{Umbrella}|\text{Rain}, \text{Traffic})$
- With conditional independence assumption:

  $P(\text{Traffic}, \text{Rain}, \text{Umbrella}) =$

  $P(\text{Rain})P(\text{Traffic}|\text{Rain})P(\text{Umbrella}|\text{Rain})$
- *Bayesian networks* will help us to express conditional independence assumptions